

INSTITUTE OF SCIENCE & TECHNOLOGY
B.Sc. (HONS) IN CSE PART-I, SECOND SEMESTER, MODEL TEST-2022
CSE-510221
(Digital System Design)

Marks

1. (a) What is binary number? How can any binary number can be converted into its equivalent decimal number and vice-versa?

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- (b) Describe the main functional units of a digital computer.

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2. (a) Simplify the expression

$$Z = \bar{P}\bar{Q}\bar{R} + \bar{P}\bar{Q}R + \bar{P}Q\bar{R} + \bar{P}QR + PQR$$

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- (b) Design a logic circuit with inputs A, B, C so output Z is HIGH whenever A is 0 or whenever A = C = 1.

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- (c) Determine the output for the following circuit: --

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